Problem Statement#1: Automated Academic Timetable Generation System

Develop a web-based system that allows the timetable coordinator to enter departmental teaching requirements, available classrooms and laboratories, faculty availability, and university working hours. The system must automatically generate a timetable that satisfies these requirements and constraints.

1. Coordinator inputs

Input category	**Required details**
University timings	Working days, official start and end times, period duration, and scheduled breaks.
Classrooms and laboratories	Room number/name, type, seating capacity, and availability.
Programmes and student groups	Programme, semester, section, student strength, and practical batches with their strengths.
Subjects and practicals	Subject code/name, programme and semester, theory or practical, and assigned faculty.
Weekly teaching requirements	Number of theory periods per week, number of practical sessions per week, and consecutive periods required per practical session.
Faculty constraints	Available working days and time ranges, unavailable slots, and any fixed teaching commitments.
Additional scheduling constraints	Fixed sessions, required laboratory type, and any limit on daily teaching periods.

The system must distinguish between **mandatory constraints**, which cannot be violated, and **preferences**, such as preferred teaching slots, which should be accommodated where possible.

The generated timetable must satisfy these mandatory rules:

1. All sessions fall within university working hours and avoid scheduled breaks.
2. Faculty members are scheduled only during their available times.
3. No faculty member, room, or student group has overlapping sessions.
4. A whole-class session cannot overlap with a practical session involving any students from that class.

5. Room capacity meets the strength of the assigned class or practical batch.
6. Practical sessions use a suitable laboratory and occupy the required consecutive periods.
7. Every subject receives its specified weekly theory periods and practical sessions.
8. Fixed teaching commitments remain in their assigned slots.

3. Handling an impossible timetable

If the entered requirements cannot be accommodated, the system must clearly report the scheduling problem instead of presenting an incomplete timetable as successful.

For example:

> "Subject DFIS103 requires three weekly periods, but its assigned faculty member is available for only two periods."

The system should identify unassigned sessions and relevant constraints so the coordinator can revise the inputs and generate the timetable again.

4. Review and publication

The coordinator must be able to:

- View timetables programme-wise, faculty-wise, and classroom/laboratory-wise.
- Manually adjust generated sessions, with immediate conflict validation.
- Lock selected sessions and regenerate the remaining schedule.
- Save a draft and publish a validated timetable.
- Print or export the timetable.

5. Example requirement

For **M.Sc. DFIS, Semester I**, the coordinator enters:

- Student strength: **40**, divided into **two practical batches of 20**.
- A subject requiring **three theory periods per week**.
- A practical requiring **one session of two consecutive periods per batch per week**.
- Assigned faculty availability and suitable classrooms/laboratories.
- University working days, hours, and breaks.

The system must automatically schedule all three theory periods and both batch practical sessions while respecting faculty availability, room capacity, and every overlap constraint.

Expected outcome: A usable automatic scheduling system in which the coordinator defines the resources, teaching requirements, and constraints, then reviews and publishes the generated timetable.